

A PROJECT PHASE – I REPORT ON

**Transparency of Ethical Product's and its
source in Supply Chain**

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THE REQUIREMENTS
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Bachelor Of Engineering (Information Technology)

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CERTIFICATE

This is to certify that the project based seminar report entitled “**Transparency of Ethical Product’s and its source in Supply Chain** ” being submitted **Vaishnavi Pawar (BIT52), Sumedha Bhagwat (BIT54), Rutuja Awade (BIT55), Priya Yadav (BIT57)** is a record of bonafide work carried out by him/her under the supervision and guidance of **Mrs. Preeti Patil** in partial fulfillment of the requirement for **BE (Information Technology Engineering)** from Dr. D. Y. Patil Institute of Technology, Pimpri, Pune in the academic year 2020-2021.

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ABSTRACT

As there is an enormous progress in the Information Technology field, there is also a rise in the economic which in turn creates a competitions between the companies. In the Blockchain era we have entered the Supply Chain Management System. Compared to the traditional Supply Chain Management System the Supply Chain Management using Blockchain is more effective and useful in its own ways, it also surpasses the limitations of the traditional supply chain management.

Supply chain acts as a primary concern in the business field while transporting of the products among the producer and consumer in the right and systematic way. Many companies across the globe face many challenges to manage the supply chain from source to destiny. We have created a system which maintains a transparency of products and its source in Supply Chain using Blockchain. The objective of this system is the development of an integrated information system in which the digital stream of data generated in the supply chain of products can be assigned to these products. That's the reason why the majority of companies nowadays approach this concept. There are significant advantages of adopting Blockchain in supply chain management, they are: Customer Fulfilment, Anti-Counterfeit, Identify the problems, cost efficiency and rise in Economy.

Key Words: Blockchain, Supply chain, source, transparency, products, Counterfeit product, QR code

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LIST OF ABBREVIATIONS

ABBREVIATIONS	ILLUSTRATIONS
UML	Unified Modeling Language
DFD	Data Flow Diagram
PBFT	Practical Byzantine Fault Tolerance
SCP	Stellar Consensus Protocol
FBA	Federated Byzantine Agreement
BFT	Byzantine Fault Tolerance

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO PROJECT

Blockchain is a new type of data system that maintains and records all the details regarding to the field in a way that permits multiple stakeholders to confidently share access to the same data and information.

A blockchain is a type of distributed ledger technology (DLT), which means it is a data ledger that is shared by multiple entities operating on a distributed network. The Blockchain technology operates by recording and storing every transaction across the network in a cryptographically linked block structure that is replicated across network participants. Each block in the blockchain has a hash, which is the output of an algorithm that turns the contents of the block into a random mix of letters and numbers. By mathematically validating that the hashes match the expected values, users can trust that the data has not been tampered with. Relevant actors or organizations participating in the distributed network can serve as “nodes” to participate in consensus, a process that keeps each blockchain node in sync and handles the addition of new blocks. There are different mechanisms that could be used to establish consensus that have trade-offs between confidentiality, throughput, and security, and vary depending on the implementation and use case.

There are different types of Blockchains:

- Public Blockchain,
- Private Blockchain
- Consortium Blockchains
- Hybrid blockchains

Depending on the type of blockchain platform being used, blockchains can be designed to provide different levels of access to the data on the blockchain (“on-chain” data). This means that it can provide an increase in transparency of the data between the consumer, while upholding privacy where needed. For instance, a blockchain could enable patients to regulate their own health data and choose who can access their health data.

Using blockchain during this way establishes trust in the validity of the information and improves the potential to share the info across silos while keeping sensitive data protected.

They use labelled datasets to train and to render it accurate by changing the parameters of the learning rate. After that, they apply parameters of learning rate to the dataset, the techniques that implement supervised learning such as multilayer perceptron (MLP) to build the model based on the history of the database.

1.2 MOTIVATION BEHIND PROJECT TOPIC

In complex production, a large amount of data is generated which can be the source for value-add for actors in the value chain. Blockchain-based systems provide the potential to capture information along the production chain, to provide means for collaboration and they ensure integrity of information.

The worldwide improvement of an item or innovation consistently accompanies hazard factors, for example, forging and duplication also, use of harmful products. Due to awareness of cruelty free environment most users prefer to know the source of the products they are using.

Forging items can influence the organization's name and the client's wellbeing. Now-a-days discovery of forged item is the greatest test. Counterfeit and Environment harmful products are causing a significant impact on the organization and the client's wellbeing.

Hence, the creators of products are defying enormous misfortune. India and different nations are battling to create an environment friendly product.

1.3 AIM AND OBJECTIVE(S) OF THE WORK

1.3.1 Project Aim:

To establish a reliable, responsible, modernized, transparent, and secured supply chain against counterfeiting attacks. To ensure integrity of information and to benefit in revenue growth which will help in Economy development.

1.3.2 Project Objectives:

The objectives of this project are:

- The objective is to develop an integrated information system in which the digital stream of data generated in the supply chain of products can be assigned to these products.
- To Design a system using Blockchain to overcome Counterfeit of products.
- To secure the details of the product using a QR code.
- Provide security and trust between the customer and manufacturer by offering data to customer.
- No manipulation of the stored data can be done.
- Furthermore, the information should enable traceability of product, its sub-products and all steps in manufacturing with their associated quality parameters

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION:

Here we present a review on tracing the source of the product in the supply chain. Due to the traditional supply chain Counterfeit products are growing exponentially with the enormous amount of online and black-market. So, there is a strong need to address the challenges of detecting the source of the products and designing appropriate technology to improve detection accuracy. It is one of the active research areas to be explored in the current world.

2.2 RESEARCH PAPERS:

- Design of an efficient image protection method based on QR code (IEEE-2021):

In this paper, the author proposed a technique to protect image data based on QR codes and describe the design and verification results. In the aspect of image protection, the technology presented in this paper can enhance security through the creation of a secure image with both de-identification and encryption applied. In terms of QR code, our proposed method optimizes data to meet capacity limits and to eliminate security vulnerabilities.

- A Blockchain-Based Application System for Product Anti-Counterfeiting (IEEE-2021): This paper is the first Blockchain system that proposes a fully functional anti-product forgery system. By paying a very low transaction fee, users of our system no longer need to be concerned about the possibility of acquiring a counterfeited product.

- Fake Product Detection Using Blockchain Technology (IEEE-2021): In this paper, the author proposed a system which is useful for the customer to detect fake

products in the supply chain. Customers are able to scan QR codes which is assigned to the product and can get all the information like transaction history, current owner based on which end-user can check whether the products they bought are authentic or not.

- An Efficient Bar/QR Code Recognition System for Consumer Service Applications (IEEE-2020): This paper presents an image processing-based method for automatic analysis of Bar/QR codes. The proposed method involves the filtering of the images which removes the noises associated in the images. Further, the images are converted into grey scale and binarization is performed. The experiments conducted using this yielded 87% accuracy
- A Blockchain-based Supply Chain Quality Management Framework (IEEE-2020): In this paper, the author proposed a blockchain-based framework. The framework comes up with a theoretical basis for brilliant quality management of the supply chain which is based on the Blockchain Technology. Furthermore, it imparts a foundation to develop theories related to data resource management in distributed, virtual organizations.
- Smart Tags for Brand protection and anti-counterfeiting in the wine industry (IEEE-2020): This paper describes a brand protection and anticounterfeiting solution for the wine industry based on smart tags and Cloud enabled technologies. The main idea utilized in this technique are the usage of smart tags which are used to utilize quick response codes and functional inks supported by the Cloud system and which maintains a two-way communication between the winemaker and the end-user.

2.3 COMPANY'S STANCE TOWARDS BLOCKCHAIN:

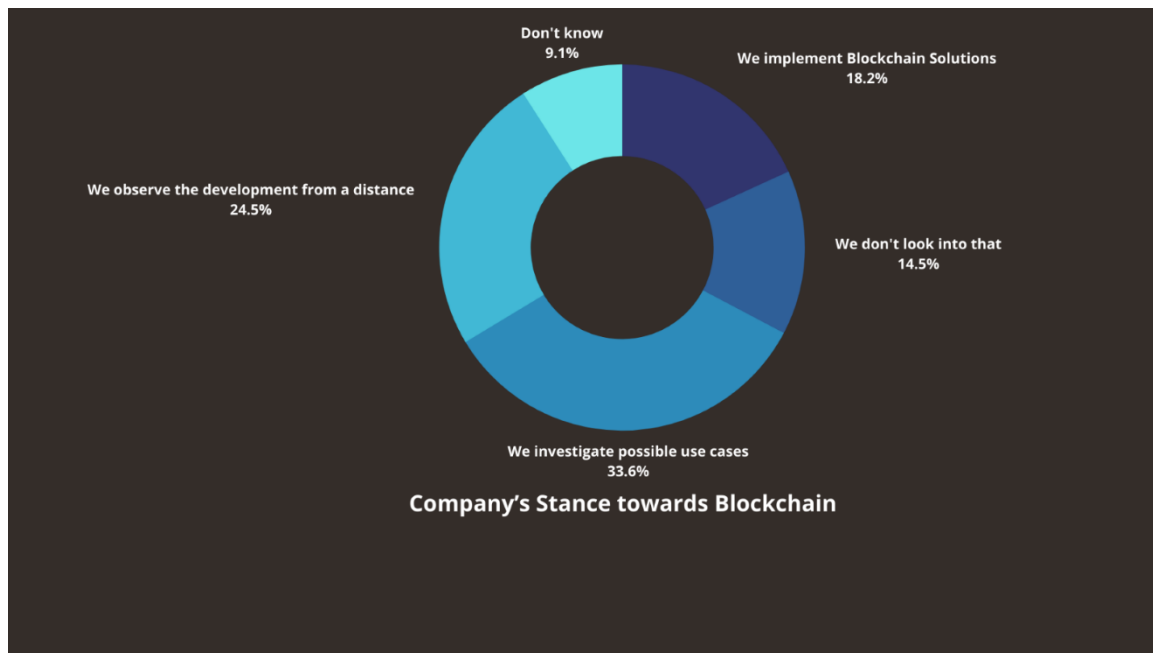


Figure 2.3.1 Company's Stance towards Blockchain

The above pie chart gives us an idea of the company stance towards blockchain technology. Around 24.5% companies observe the development from a distance and do not use the technology. 33.6 % companies investigate the possible use cases. 14.5% companies are not interested in the technology. While 18.2% have successfully implemented the blockchain technology.

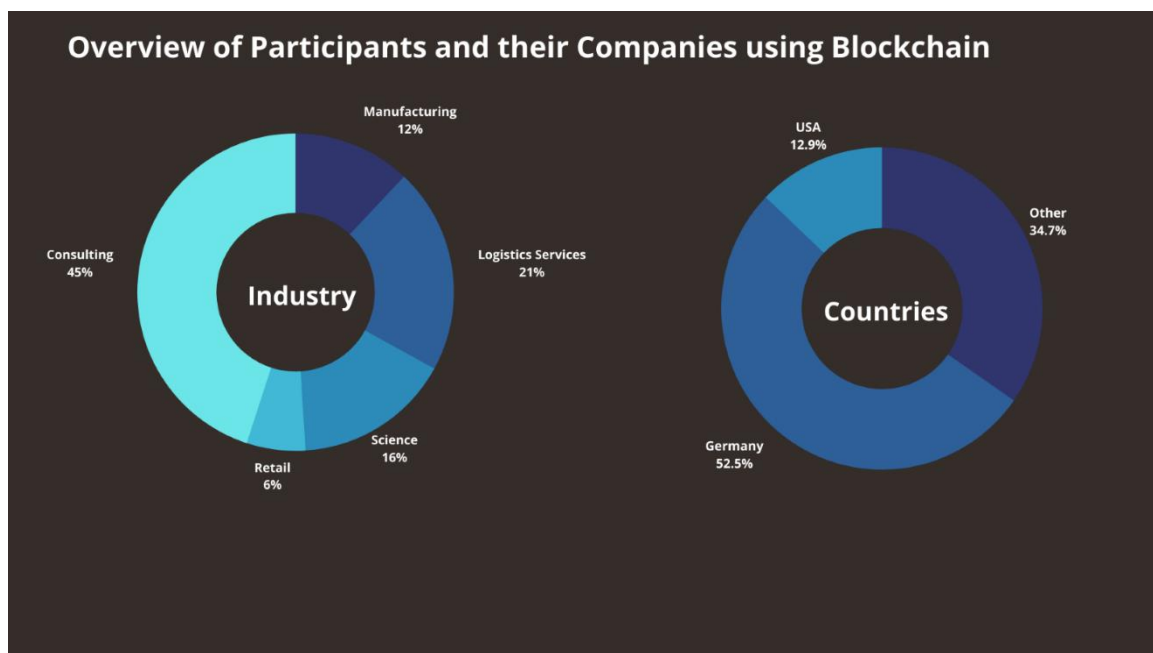


Figure 2.3.2 Overview of Participant's and their Companies using Blockchain

The above pie chart gives the overview of the participants and their companies using blockchain technology.

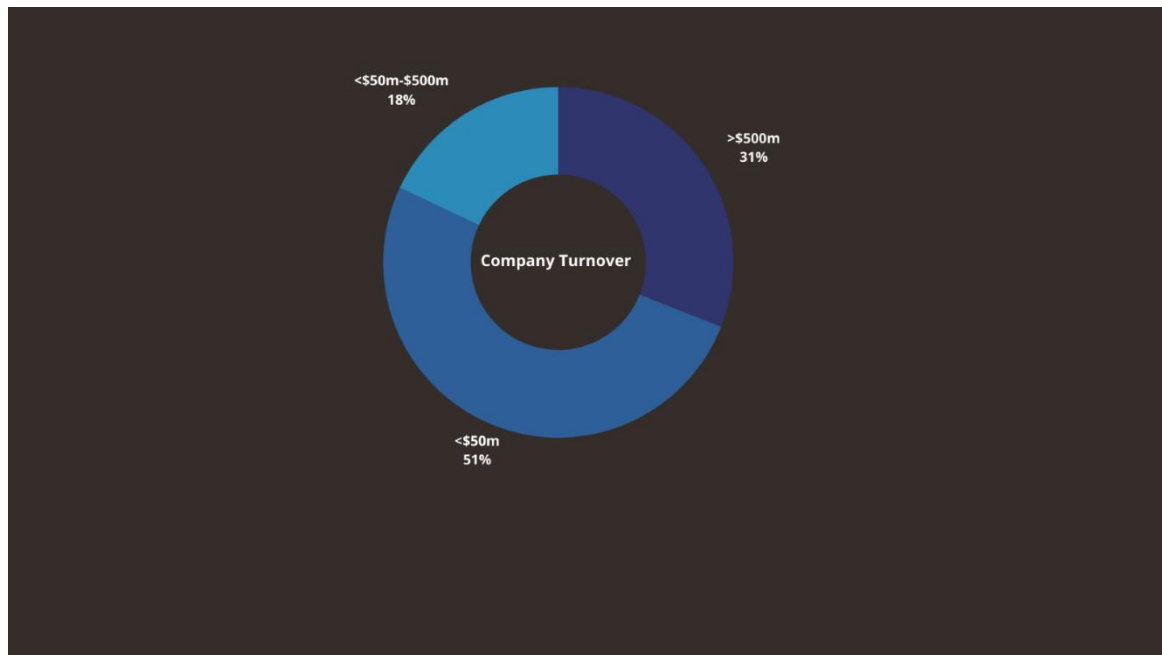


Figure 2.3.4 Overview of Companies turnover using Blockchain

The above pie chart gives the overview of the companies turn over using block chain technology. This pie chart shows us that Blockchain is a major contributor in majority of the companies with a turn over of above 550 million.

CHAPTER 3

EXISTING SYSTEM & PROPOSED METHODOLOGY

3.1 EXISTING SYSTEM:

Traditional Supply Chain Management mechanisms usually suffer from a wide scope of issues such as lack of information sharing, long delays for data retrieval, and unreliability in product tracing. There is no existing work in literature that provides an end-to-end solution for supply chain.

The current Supply chain system doesn't share the same data between independent actors and different actors without the actors having to directly interact and build a direct relationship with one another.

The existing business models include various processes associated with sending, validating, and reconciling data among different parties, where there is no security in the data, where data can't be shared across multiple actors, also there is also a need to revalidate and reconcile data which in terms affect the economy.

The first problem that we address in this report is the centralized anti-counterfeiting supply chains. Despite their potential to identify the forged products, they still experience single point processing, storage, and failure. Furthermore, they do not offer transparency in supply chain as they do not allow the supply chain nodes to verify the authenticity of a product's data.

3.2 PROPOSED METHODOLOGY:

Blockchain is more than just a technology. It is a critical enabler of innovation as it acts as a catalyst for changing the way that existing trading partners in a supply chain work together.

Blockchain introduces a means to share data across multiple, possibly mistrustful, parties without the necessity for an intermediary, reducing the need to revalidate and reconcile data.

The proposed solution uses Smart contracts to assure an efficient, secure and trusted environment for the supply chain activities. The main contributions of the proposed system are as follows.

- It serves as an end-to-end distributed supply chain system that introduces a traceability scheme and trading and delivery mechanisms, a reputation

system for credibility assurance of the entities and an autonomous transaction system.

- It achieves the desired properties of accountability, credibility, auditability, autonomy and authenticity
- It acts as a better alternative to the existing Traditional supply chain systems enabling a scalable and auditable system. It presents the algorithms of smart contracts and evaluates them for vulnerability and their costs of products in the network.
- It is resilient against well-known attacks and provides inherent important security features. We perform a detailed vulnerability assessment of our system and discuss how our system is robust and ensures security against malicious attacks.

In the proposed framework, the framework produces QR codes utilizing Blockchain innovation. This innovation stores exchange records in blocks. These squares are secure and difficult to access and change the data from it. By utilizing a QR code we can recognize the products source and also detect whether it is fake or not by using a smartphone.

By implementing this system, the transportation of the goods will be done with their real price, facilitates the planning, decreases the expenditures and increases the customer satisfaction.

CHAPTER 4

PROJECT PLAN

4.1 TIME LINE OF THE PROJECT:

The development of the project started with two phases in mind. The first phase consists of research across various domain to select the topic which will be beneficial to the society. The research part of the project after the selection of topic took around 2-3 months. The second phase is the implementation phase which is still in process. The complete implementation of the project will be done in upcoming 3-4 months.

4.2 FEASIBILITY STUDY:

In figure 1.1, 1.2, 1.3 which states the overview of participants and companies those who use block chain technology. The companies using it experiences a huge turnover. This pie chart shows us that Blockchain is a major contributor in majority of the companies with a turnover of above 550 million.

4.3 RISK ANALYSIS:

Blockchain is an emerging technology which is still in the proof-of-concept phase, there are also some of blockchain types who have not yet proven themselves in an industrial environment consequently real threats and challenges will not appear in the few newt years of its implementation. This implementation must be carefully analysed and followed before being adopted because any little mistake in implementation would have major risks and bad consequences.

4.4 EFFORT:

The functionalities and architecture of the prototype were designed and developed in an iterative process. Several workshops with 2 - 4 persons were held and aspects of different architectural approaches were discussed.

4.5 NUMBER OF PEOPLE:

- Vaishnavi Pawar, BIT52
- Sumedha Bhagwat, BIT54
- Rutuja Awade, BIT55
- Priya Yadav, BIT57

CHAPTER 5

PROJECT REQUIREMENTS SPECIFICATIONS

5.1 Programming language: Python, Java

5.2 Software Requirement:

- Windows 7 or higher.
- SQL 2008
- Visual studio 2010
- Android studio IDE
- Firebase Cloud

5.3 Hardware Requirement:

- Processor – i5
- Hard Disk – 5 GB
- Memory – 1GB RAM
- Internet Connection

CHAPTER 6

ARCHITECTURE OF THE SYSTEM

6.1 ARCHITECTURE:

The below figure represents the Architecture of our system.

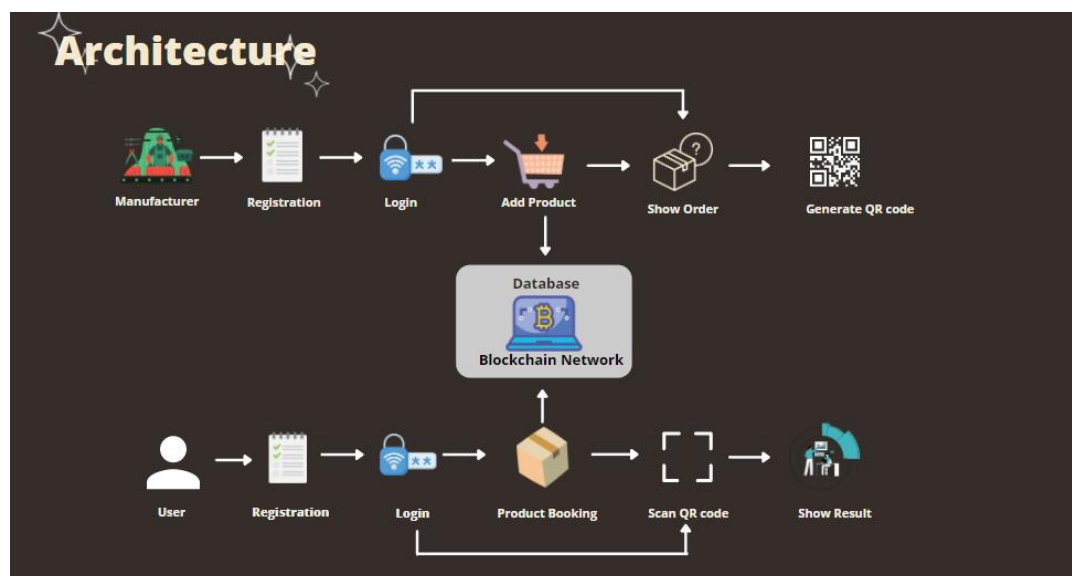


Figure 6.1.1 Architecture of the System

We proposed a system using blockchain technology as an android application for the tracing of products back to its source and for detection of counterfeit products. The proposed system ensures that the products bought are safe and sustainable right from its source which are used in day-to-day life. The proposed system consists of three main parts, customer or user android application, Manufacturer's or company's android application, and Cloud/ Database.

The first application is the Manufacturers or company side application in which we have to first register ourselves. After registration login into the application, we have some options. One option is to add a product in which the manufacturer can add the product details. Another option is to show the order in which they can see customers' order details and after that, they can decide to accept or reject the order. The manufacturer also can see the product is delivered or not.

A second application is the Customer application in which we have to first register in-app after that we can log in to the application using id and password. In this application, there is an option to show products where customers can see the product details like name, total quantity, price of a product, details of the manufacturer. In that, we can product book the product by inserting the quantity of product.

In this application, we can see orders using show my order where we can see the product details, name, quantity, date, time, price, and status of product which is produced is delivered or not. In this app we have a QR code scanner in which we scan the QR code of the product then it shows the source of the product. There is another option which is a blockchain in which it displays the name of generated block product quantity, generated Hash Value, and the product is corrupted or not.

In this project there is a customer login, in the application. After login, he fills in the details for ordering the product and book the product. The order of the product can be shown to the manufacturer. Manufacturer deciding whether the product request is acceptable or not. After the manufacturer accepting the order of the product it generates the unique QR code of a product. Once an order of product is stored on the network hash code is generated of that product and it is possible to maintain the transaction of the product. In the proposed system QR code is generated for a particular product. Customers scan the QR code on the product or package using the smartphone's QR code reader application or customer application have the option of QR code scanner.

After scanning, we get the result of the product source. In the end, the Blockchain system holds these product details along with a history of transactions to enable the tracking of the product along the distribution chain. All the product details, block name, a hash value is stored in the firebase cloud database

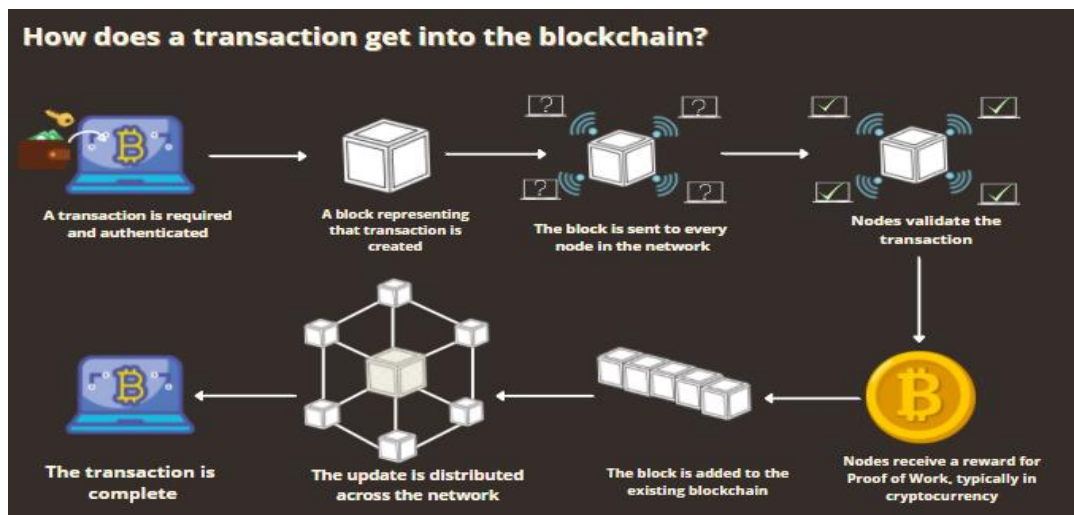


Figure 6.1.2 How does a transaction get into the blockchain

Fig 6.1.2 shows how a transaction gets into the Blockchain. When a transaction is requested and authenticated through the platform a new block is created. The Blockchain technology operates by recording and storing every transaction across the network in a cryptographically linked block structure that is replicated across network participants. The block which is created regarding the transaction is sent to every node in the network. Nodes validate the transaction. Typically, in Cryptocurrency it is noticed that nodes i.e., participants receive a reward for proof of work. Each block in the blockchain has a hash, which is the output of an algorithm that turns the contents of the block into a random mix of letters and numbers.

By mathematically validating that the hashes match the expected values, users can trust that the data has not been tampered with. Relevant actors or organizations participating in the distributed network can serve as “nodes” to participate in consensus, a process that keeps each blockchain node in sync and handles the addition of new blocks. The block is then added to the existing blockchain and the update regarding it is distributed across the network. Hence the transaction is completed.

CHAPTER 7

HIGH LEVEL DESIGN OF THE PROJECT (UML/DFD)

7.1 UML DIAGRAM:

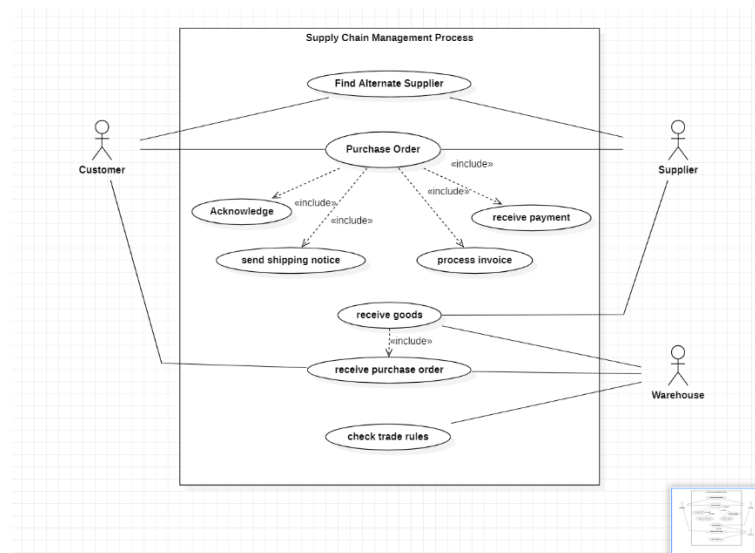


Figure 7.1.1 UML diagram

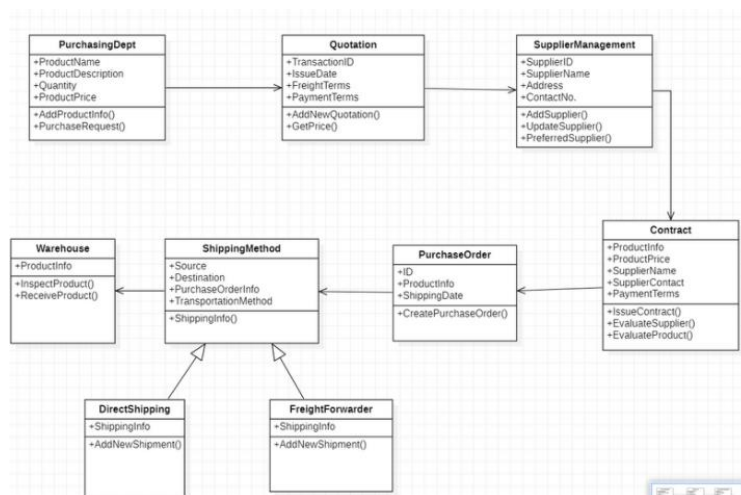


Figure 7.1.2 UML diagram 2

7.2 DFD Level 0:

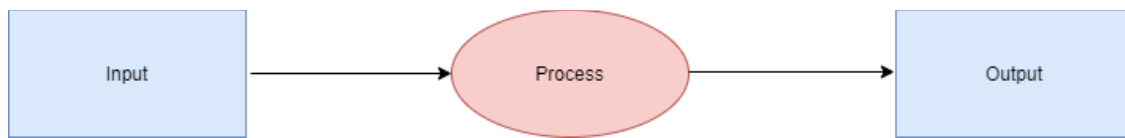


Figure 7.2.1 DFD Level 0

7.3 DFD Level 1:

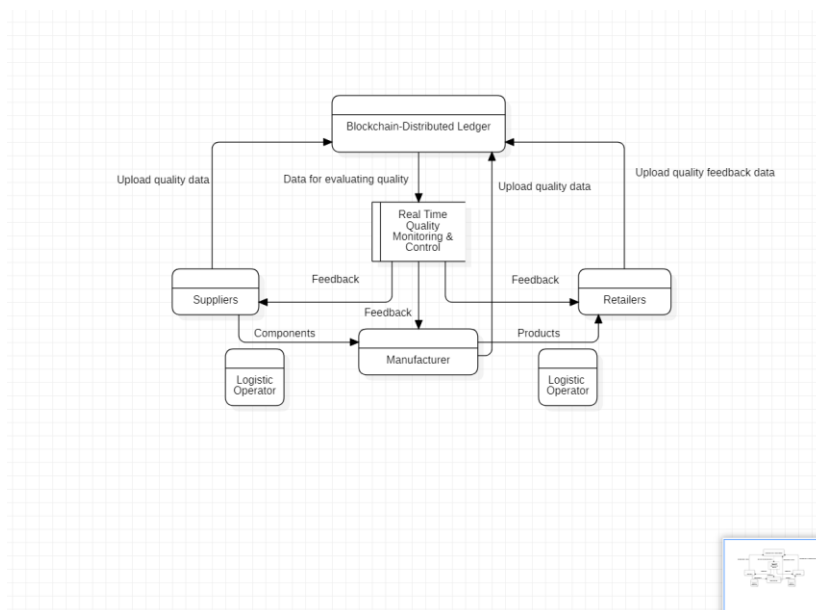


Figure 7.2.2 DFD Level 1

CHAPTER 8

PROTOCOLS

8.1 CONSENSUS PROTOCOLS:

- **PBFT (Practical Byzantine Fault Tolerance):** PBFT is a replication algorithm that can tolerate Byzantine faults. PBFT progresses through a series of views. Each view has a primary node (i.e., proposer,) which is selected in round-robin order. The other nodes (replicas) in the view are called backups. A client sends a request to the primary. The primary multicasts a signed pre-prepare message for this request to the backups. The backups accept the pre-prepare message, and broadcast signed prepare messages. If the backups receive $2f$ prepare messages for the request (where f is the maximum number of replicas that may be faulty), they multicast signed commit messages. Backups execute a view change protocol, in case of a faulty primary. PBFT has the following issues. First, changing the view is subtle and a bit complicated. Second, all previous client requests since the last commit are migrated to the new view.
- **Tendermint:** Tendermint delivers security for replicating an application on multiple nodes as it can work even if up to one-third of nodes in the network fail in arbitrary ways. Tendermint is a consensus protocol that does not include proof-of-work mining, which overcomes the energy and resources consumption issues, and speeds up blocks' validations. Tendermint is based on PBFT, and it involves three stages of voting to reach consensus (propose, prevote, and precommit). A proposer proposes a new block, then the validators prevote on the block and only proceed to precommit if they receive more than $2/3$ of prevotes. Validators only accept the block if more than $2/3$ of precommits are received. Tendermint requires a fixed known set of validators. Voting on a block proceeds in rounds, where each round has a new proposer. The validators vote on whether to commit the block or advance to the next round. Tendermint is notable for its simplicity, performance, and fork-accountability.
- **Hyperledger Fabric:** Hyperledger Fabric employs PBFT as its consensus algorithm [5]. Thus, it can tolerate up to $1/3$ byzantine nodes in a blockchain network. In Fabric v0.6, there exist a fixed number of validation peers responsible for executing the consensus protocol. A proposer can submit a transaction to any

of them. Then, the chosen peer broadcasts this transaction to the other peers. One of the validation peers is selected as a leader. When generating a block, the leader broadcasts it to all peers. When a validation peer receives this block, it hashes it, broadcasts the hash to all other peers, and begins counting their responses. If two-thirds responses were received with the same hash, it commits the new block to its local ledger. Hyperledger Fabric, like Tendermint, suffers partial centralization since it employs a fixed known number of validation peers.

- **Stellar Consensus Protocol (SCP):** SCP is a consensus protocol that utilizes quorums, where a quorum is a set of nodes from a network sufficient to reach an agreement. SCP is based on Federated Byzantine Agreement (FBA), in which SCP exploits the concept of a quorum slice. A quorum slice is the subset of a quorum that can cause one particular node to reach an agreement. The key idea in FBA is that every node chooses its own quorum slices. A node accepts a vote or a transaction when a threshold (e.g., 2/3) of nodes in its quorum slice confirm it. However, SCP requires the quorum slices to overlap. The previously discussed protocols such as PBFT, Hyperledger Fabric, and Tendermint employ a fixed and globally known set of nodes to reach a consensus. In contrast, SCP gives each node a choice to select one or more quorum slices, each of which might have different nodes. Despite the beauty of this design, it might result in undermining the consensus as quorum slices might not overlap. It is worth mentioning that we only considered the BFT (Byzantine Fault Tolerance) protocols in this section since they are more relevant to our proposed protocol.

CHAPTER 9

SHA-256 ALGORITHM

9.1 INTRODUCTION OF SHA-256 ALGORITHM:

SHA 256 is a part of the SHA 2 family of algorithms, where SHA stands for Secure Hash Algorithm. Published in 2001, it was a joint effort between the NSA and NIST to introduce a successor to the SHA 1 family, which was slowly losing strength against brute force attacks.

The significance of the 256 in the name stands for the final hash digest value, i.e. irrespective of the size of plaintext/cleartext, the hash value will always be 256 bits.

The other algorithms in the SHA family are more or less similar to SHA 256. Now, look into knowing a little more about their guidelines.

9.1.2 Characteristics of SHA-256 algorithm:

Some of the standout features of the SHA algorithm are as follows:

- **Message Length:** The length of the cleartext should be less than 264 bits. The size needs to be in the comparison area to keep the digest as random as possible.
- **Digest Length:** The length of the hash digest should be 256 bits in SHA 256 algorithm, 512 bits in SHA-512, and so on. Bigger digests usually suggest significantly more calculations at the cost of speed and space.
- **Irreversible:** By design, all hash functions such as the SHA 256 are irreversible. You should neither get a plaintext when you have the digest beforehand nor should the digest provide its original value when you pass it through the hash function again.

9.2 Applications of SHA-256 algorithm:

- **Digital Signature Verification:** Digital signatures follow asymmetric encryption methodology to verify the authenticity of a document/file. Hash algorithms like SHA 256 go a long way in ensuring the verification of the signature.
- **Password Hashing:** As discussed above, websites store user passwords in a hashed format for two benefits. It helps foster a sense of privacy, and it lessens the load on the central database since all the digests are of similar size.
- **SSL Handshake:** The SSL handshake is a crucial segment of the web browsing sessions, and it's done using SHA functions. It consists of your web browsers and the web servers agreeing on encryption keys and hashing authentication to prepare a secure connection.
- **Integrity Checks:** As discussed above, verifying file integrity has been using variants like SHA 256 algorithm and the MD5 algorithm. It helps maintain the full value functionality of files and makes sure they were not altered in transit.

CHAPTER 10

GUI



Figure 10.1 working module

In this project there is a customer login, in the application. After login, he fills in the details for ordering the product and book the product. The order of the product can be shown to the manufacturer. Manufacturer deciding whether the product request is acceptable or not. After the manufacturer accepting the order of the product it generates the unique QR code of a product.

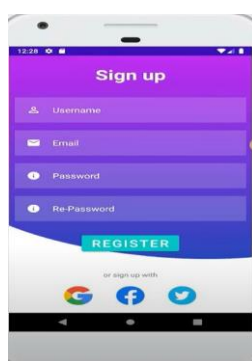


Figure 10.2 working module

Once an order of product is stored on the network hash code is generated of that product and it is possible to maintain the transaction of the product. In the proposed system QR code is generated for a particular product. Customers scan the QR code on the product or package using the smartphone's QR code reader application or customer application have the option of QR code scanner.

CHAPTER 11

APPLICATION OF BLOCK CHAIN TECHNOLOGY

11.1 APPLICATIONS OF BLOCKCHAIN TECHNOLOGY:

Blockchain technology has made significant strides in its development and widespread adoption in recent years, and there's no sign of it slowing down. In fact, according to Deloitte's 2021 Global blockchain development Survey, almost 76% of executives surveyed said they expect digital assets will be a solid alternative to fiat currencies for global finance in the next 5–10 years.

Below are some applications of Blockchain Technology: -

1. **Faster Transactions:** - Due to the secured & decentralized nature of blockchain technology, it's incredibly difficult (if not impossible) for hackers or nefarious parties to tamper with transactions. Blockchain-verified data is highly secure and trustworthy, meaning transactions can be processed much faster than in today's world without compromising security.
2. **Digital Identity:** - Today, we use passwords and authentication questions to prove who we are online. Blockchain could replace this system with a digital identity that is safe, secure, and easy to manage.
3. **It'll Be Easier to Treat Health Problems:** - As we mentioned earlier, digital identities could be stored and managed through blockchain technology and this goes for health records as well.

As you can see, there's a ton of potential for blockchain technology and it's constantly expanding. Also, the future of blockchain technology looks bright, and when you consider that it's already showing promise in almost every industry, it seems like the best is yet to come.

Industry Segment	Uses for Blockchain
Government	• Voting • Taxes • Tender processes
Industrial	• Manufacturing processes • Internet of Things (IoT) device management
Financial Services	• Foreign exchange • Corporate debt / bonds • Trading platforms • Payment remittance
Insurance	• Claims processing • P2P insurance • Ownership titles • Sales and underwriting
Retail	• Loyalty points • Identity management • Trusting industry • Capital asset management • Letters of credit

Figure 11.1.1 Applications of Blockchain in different sectors

11.2 ADVANTAGES OF BLOCKCHAIN TECHNOLOGY:

- Producers will have security. They will be able to receive a reasonable percent of profit. They will have knowledge regarding the various factors affecting their crops (i.e., unprecedented seasonal changes, soil quality, water availability, etc.) and thus will be able to take well informed decisions.
- Traders can eliminate competition.
- Processors will be able to procure goods they specifically want for the product they are trying to produce and also control the quality from the bottom up.
- Traceability: Players in different tiers can be held responsible for any damages.
- Increase in multidirectional information transfer.
- Overall increase in efficiency: Only required amount of goods will be produced thus decreasing wastages and increasing profits.

- Using blockchain in this way establishes trust in the validity of the data and improves the ability to share the data across silos while keeping sensitive data protected.
- Blockchain introduces a means to share data across multiple, possibly mistrustful, parties without the necessity for an intermediary, reducing the need to revalidate and reconcile data.

11.3 DISADVANTAGES OF BLOCKCHAIN TECHNOLOGY:

- **Blockchain is not a Distributed Computing System:** - Blockchain is a network that relies on nodes to function properly. The quality of the nodes determines the quality of the blockchain. For example, Bitcoin's blockchain is strong and incentivizes the nodes to participate in the network. However, the same cannot be true for a blockchain network that does not incentivize the nodes.
- **Scalability Is an Issue:** - Blockchains are not scalable as their counterpart centralized system. If you have used the Bitcoin network, then you would know that the transactions are completed depending on the network congestion. This problem is related to scalability issues with blockchain networks.
- **Blockchain Cannot Go Back — Data is Immutable:** - Data immutability has always been one of the biggest disadvantages of the blockchain. It is clear that multiple systems benefit from it including supply chain, financial systems, and so on.
- **Cost And Implementation Struggle:** -The underlying cost of implementing blockchain technology is huge. Even though most of the blockchain solutions including Hyperledger are open source, they require a lot of investment from the organization that is willing to pursue it.

11.4 LIMITATIONS OF BLOCKCHAIN TECHNOLOGY:

Lack of Awareness: -

There is a lot of discussion about blockchain, but people do not know the true value of blockchain and how they could implement it in different situations.

Limited availability of technical talent: -

Today, there are a lot of developers available who can do a lot of different things in every field. But in the blockchain technology, there are not so many developers available who have specialized expertise in blockchain technology.

Immutable: -

In immutable, we cannot make any modifications to any of the records. It is very helpful if you want to keep the integrity of a record and make sure that nobody ever tampers with it. But immutability also has a drawback. We can understand this, in the case, when you want to make any revisions, or want to go back and make any reversals.

Key Management: -

As we know, blockchain is built on cryptography, which implies that there are different keys, such as public keys and private keys. When you are dealing with a private key, then you are also running the risk that somebody may lose access to your private key.

Scalability: -

Blockchain like bitcoin has consensus mechanisms which require every participating node to verify the transaction. It limits the number of transactions a blockchain network can process.

Consensus Mechanism: -

In the blockchain, we know that a block can be created in every 10 minutes. It is because every transaction made must ensure that every block in the blockchain network must reach a common consensus

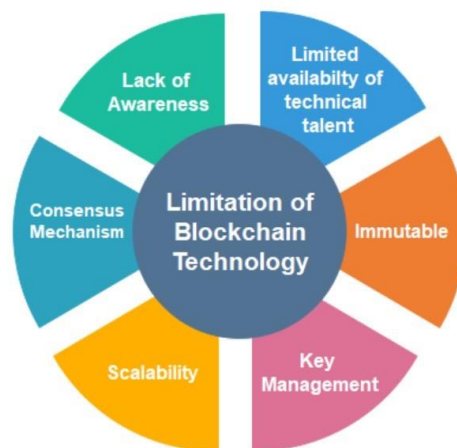


Figure 11.4.1 Limitations of Blockchain Technology

CHAPTER 12

FUTURE SCOPE

We believe that ultimately global trade will evolve into a battle of supply chains, where efficiency, block chain optimization, and innovative financing will yield best in breed organizations. As blockchain gains momentum, “there will be real-time visibility, reduced complexities, improved accuracy and efficiency in the system, which in turn will help reduce costs and enhance trust.

Its usefulness will not be limited to shippers or customers but will extend to lowering trade barriers, especially for developing nations which are not able to participate in global trade owing to cost considerations”.

For example, smart contracts can be used for transactions to exchange money, property, shares, or anything of value where blockchain defines the rules and penalties of the contract and automatically enforce those obligations, including expiration dates and validity.

Additionally, blockchain can be integrated in the Internet of Things device management, voting confidence, and P2P insurance. All of these applications of blockchain are already being vetted and show favourable results.

We aim to address some of these big issues as a future job and build alternatives to them. We also intend to incorporate automated payments and delivery proof into our suggested solution-whereby parties are paid in a centralized and efficient manner using cryptocurrencies through smart chances for effective physical plant and product distribution.

The golden rule which states “those who have the gold, rule” seems probable as access to working capital will become necessary in a more transparent environment. Overall, “when adopted at scale, the solution has the potential to save the industry billions of dollars and benefit global trade and the world economy”.

CHAPTER 13

CONCLUSION

Blockchain makes it possible for a system of independent actors to share and trust a record of digital assets, transactions, and information. In doing so, it offers the potential to disrupt and transform existing business models. When a blockchain-based traceability system for the full end-to-end supply chain for a given food commodity is not feasible, a minimum viable product should be prioritized.

The successful application of blockchain requires strong planning, mobilization, and coordination. Private, public, and hybrid blockchain solutions each have unique strengths and weaknesses depending on specific requirements. It is not necessary to build applications on a public blockchain to reap the benefits of transparency and accountability. In addition to core business drivers and financial measures of success, blockchain has potential to drive impact in social good use cases, such as environmental sustainability

The important factor in supply chain management is their information and management. The information technology can help in this connection. With use of this technology, remarkable progresses do in all parts of supply chain. In addition to accomplished operations facilitating in supply chain, use of information technology can remove many of problems which exist in this connection. It is not necessary for all supply chain data to be accessible across all actors. Actors should agree on a data model that can share required traceability data and protect sensitive product data on the blockchain. Supply chain actors may need to invest in closing the digital divide to participate.

CHAPTER 14

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APPENDIX-A

Plagiarism Checker: [Smallseotools.com](https://smallseotools.com/plagiarism-checker/)

Abstract:

RESULTS

1. Completed 100% Checked 2. Eligibility 80% Passed 3. Plagiarism 100% Passed 4. Uniqueness 100% Passed

Seriousness Wise Results | Document View | Matched Sources | Download Sources

Uniqueness	Description
Uniqueness	No shared diagram shows the full process of a transaction from a merchant website where a customer...
Uniqueness	go through a payment gateway where the merchant account (processor or processor bank) sends an...
Uniqueness	The details are provided (the payment) gateway then contacts the bank which has issued the customer's...
Uniqueness	which is then forwarded to the customer or passed for an O/I which is provided by the bank after the...
Uniqueness	and the bank then allows the transaction and releases the money for payment

100%

Completed: 100% Checked

RESULTS

0%

Plagiarism

100%

Uniqueness



Sentences Wise Result



Document View



Matched Sources

Unopen

Here we present a review on the supervised and unsupervised learning approaches used to identify the

Unopen

These are the two important categories of machine learning techniques to identify the frauds in credit

Unopen

I. Fraud Detection in Credit Card Data using Unsupervised Machine Learning Based Scheme (ICIC, 2020, 2021)

RESULTS

100% Completed 100% Checked

Progress 100%

Sentence View (html) | Document View | Metadata Source

Sentence ID	Sentence
1	In the setting context of detecting breast transposons, this tool is extended about this transposon in context.
2	For it is difficult to find if the current transposon is transcribed or not.
3	If the transcribed transposon is contained, we begin then it will not go on and to the column to be extended as well.
4	The hyperplane of database and the column of column connectivity are extremely large.
5	It is not a simple process to handle the huge amount of each efficacy.

Chapter 1

Chapter 2

Chapter 3

RESULTS

Completed: 100% Checked

Plagiarism 0% Unique 100%

Sentence Wise Result | Document View | Matched Sources

Unknow	The system logs when one starts the transaction which is made open by credit card payment, internet.
Unknow	Once the transaction starts the transaction further proceeds through a payment gateway
Unknow	A payment gateway is a non-payments service that, when integrated with the e-commerce platform,

Population	Common Variants	Rare Variants
European	99.9%	0.1%
African	99.9%	0.1%
Asian	99.9%	0.1%

The screenshot displays the 'RESULTS' tab in the IBM Security Guardium console. It features a table titled 'Nonzero User Results' with the following columns: Name, Description, Status, and Last Updated. The table contains one entry for 'IBM Security Guardium' with a status of 'Success' and a last updated time of '2023-08-01 10:00:00'. Below the table, a 'Summary' section shows a 'Total' of 1 and a 'Success' count of 1. The interface also includes a 'Combined 100% Clean' status indicator and a 'Platform: IBM' label.

Chapter 4

Chapter 5

Chapter 6

[illegible]

Unique	In this study we have prepared a method for create and find detectors
Unique	We first compared different algorithms that are used to detect, detect, and find the SVM, neural network
Unique	After comparing all the algorithms, we finally used Artificial Neural Network (ANN)
Unique	Since there is no bound in it, it shows that it can be used in any virus, algorithms, and for the first time
Unique	ANN gives more accuracy compared to any unsupervised algorithm

Chapter 7

Chapter 8